# Explain anything

Every part of Atreya, one topic per card. Each has a line you can read out word for word, and the depth underneath for when they keep asking. Find the topic, say the bold line, stop. Only go deeper if they do.

## HOW TO USE THIS AT A TABLE

| IN ONE LINE | SAY THIS | IF THEY PUSH |
|---|---|---|
| What the thing actually is. For your own head, not for saying. | Read it out verbatim. It is written to be spoken, not read. | The detail. Most judges never get here. Do not volunteer it. |

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## THE OPENER - SAY THIS WHILE THEY ARE STILL WALKING UP

SAY THIS **"At the Kumbh, fourteen people drink from a cracked pipe. They go to four different clinics at four different hours, so every doctor sees three or four cases of diarrhoea - a normal Tuesday. Nobody raises an alarm, because the number fourteen doesn't exist anywhere. We built the thing that adds it up, every five minutes."**

Then stop talking and ask them this. It hands them control, tells you which branch to take, and stops you making a speech at someone who wanted a conversation.

THEN ASK **"Do you want the thirty-second version, or the technical one?"**

**Thirty-second answer:** topics 02, 07, 08. **Technical answer:** topics 15, 16, 20. **Public-health answer:** topics 13, 12, 03. **Business answer:** topics 24, 27, 26.

## BEFORE ANYONE ARRIVES

- Laptop: **/dashboard** fronted. Tabs already open: **/replay**, **/public?s=5**. Never be seen typing a URL.
- Phone: **/entry**, PIN 1111, storage cleared so it reads "0 today".
- **/demo** -> Clear the board, then Start outbreak on sector 5. Check the alert actually appears before you relax.
- Referral card and the printed pilgrim QR poster face up on the table.
- **Redo this every 30-45 minutes.** The traffic cron keeps generating patients and the board slowly fills with real detections - correct behaviour, but it looks like noise to someone seeing it cold.

## THE FOUR RULES THAT COST YOU MOST IF BROKEN

- **Read every number off the screen, never from memory.** Say 14 when the screen shows 34 and you have lost that judge for good.
- **Hand over the phone.** A judge who used it remembers it; one who watched you use it remembers nothing.
- **Say the weakness first.** Fake data, bathing day, unverified translations. Volunteering a limitation buys more than defending one costs.
- **Stop talking when they start.** A question is interest, not an interruption.

## PART A - THE STORY

### 01 THE PROBLEM

**An outbreak is invisible not because anyone fails, but because the information never adds up anywhere.**

> SAY THIS **"Fourteen people drink from a cracked pipe. They go to four different clinics at four different hours, so every doctor sees three or four cases of diarrhoea - a completely normal Tuesday. Nobody raises an alarm, and there was no alarm to raise: the number fourteen doesn't exist anywhere. It's three here and four there, in four separate registers that get added up in November."**

The failure is not bad medicine or lazy staff. **Every one of those fourteen people received correct treatment.** The cluster only exists when somebody adds four registers together, and by then it is day four: four hundred cases, a five-hour queue, and pilgrims boarding trains home to forty districts.

The line that lands: **"Everyone got treated. Nobody turned off the tap."**

### 02 WHAT ATREYA IS

**The ten minutes after the data is entered.**

> SAY THIS **"Doctors log each visit in ten seconds on a phone that works with no network. Every five minutes we compare each area against its own recent history, at the same time of day, and flag what is genuinely unusual. The alert becomes a job a named crew has to close."**

It is not one app. It is four narrow screens, each built for one person with one job, joined by a detection engine. Nobody is asked to learn a system.

The differentiator, if you only say one thing: **"Everybody else builds the dashboard. We built the ten minutes after it."**

### 03 WHAT ALREADY EXISTED

**The 2015 Harvard Jana Swasthya project proved data collection works. It stopped there.**

> SAY THIS **"In 2015 a Harvard project ran this exact experiment at this exact festival. It proved health workers will enter the data - two thousand visits on day one across thirty-plus clinics. Then it drew charts with it. Its own published data shows antibiotics in up to thirty-one percent of encounters, recorded and never acted on."**

This matters because it removes the biggest objection before it is raised: *will anyone actually use it?* That question was answered a decade ago. The unsolved half is the loop from data to a person walking somewhere - which is the half we built.

## PART B - THE SCREENS

### 04 ENTRY FORM - /ENTRY

**Tap-only clinical entry. PIN 1111. Measured at 2.2 seconds.**

> SAY THIS **"This is the only place data comes in. Age band, sex, symptoms, diagnosis, medicine, severity - all taps, no typing. If this is slow, doctors abandon it by day three and every other screen goes dark."**

What it captures: age band (0-5, 6-17, 18-39, 40-59, 60+), sex, symptoms, diagnosis, syndrome, drugs given, severity, and risk flags. **No name, no phone number, no ID.** See topic 26.

Severity is three steps, each defined by what the clinician does rather than by how the patient feels - so two doctors pick the same one:

- **mild** - treated and sent on
- **moderate** - needs observation or a repeat visit
- **severe** - referral, IV fluids, or cannot walk unaided

There is also an antibiotic nudge: if the diagnosis does not warrant one, the form says so. It never blocks the doctor - it is a prompt, not a gate.

## 05 THE OFFLINE QUEUE

**The phone never waits for the network, and a half-failed sync loses nothing.**

> SAY THIS **"It works with no network. It queues on the phone and syncs when it can. That matters more than anything else we built."**

How it actually works, if they are technical:

- The row is written to local storage and the form clears **immediately**. The doctor never watches a spinner.
- Every record carries a **client-generated UUID**, so the phone owns the identity of the row before the server has ever seen it.
- Sync upserts with `ignoreDuplicates`. A duplicate-key error (Postgres 23505) is treated as **success** - it means the row is already safely there.
- Only the IDs actually confirmed sent are removed from the queue.

The consequence is that the same row can be sent twice, or ten times, and the database ends up with exactly one copy. That is what makes it safe to retry blindly on a bad network.

## 06 DASHBOARD - /DASHBOARD

**Twelve sectors live, updating by subscription rather than polling.**

> SAY THIS **"This is live. Twelve sectors, seventeen clinics, updating as we speak. The map is shaded by case load - and grey means a sector has stopped reporting, which is different from calm. We treat silence as a signal, not as good news."**

The grey state is worth pointing at deliberately. A sector where every clinic has gone quiet looks identical to a healthy sector if you only colour by case count - so we draw it as a blind spot instead, and name the quiet clinics underneath the grid.

Also on the screen: cases per hour against an expected line, the syndrome mix, and a stewardship panel comparing each clinic's antibiotic rate against the median.

## 07 AN ALERT

**A flagged cell, shown with the arithmetic that produced it.**

> SAY THIS **"Every alert shows its own arithmetic - observed, expected, and how many noise-widths above. An officer can argue with this. That's deliberate: they ignore anything they can't argue with."**

Click one and read the real numbers off the screen. An alert carries the sector, the syndrome, the observed count, the expected count, the score, and its tier - **watch** (dashboard only) or **act** (dispatch a crew).

An official can also attach a public advisory to an alert, which is what the pilgrim screen then shows for that area.

## 08 DISPATCH AND THE CREW - /CREW

**The alert becomes a job with a name on it.**

> SAY THIS **"The alert becomes a job with a named crew. They have to acknowledge it, and close it with what they actually found - not just mark it done."**

This is the single biggest structural difference from a dashboard. An alert nobody owns is a notification. An alert somebody must close is a task, and a task generates the timestamps that make everything downstream measurable.

Closing outcomes are specific: what was found, or explicitly "nothing found" - which is itself useful data about the false-alarm rate in the field.

## 09 TIME TO RESPONSE

**Four timestamps the system writes itself.**

> SAY THIS **"Detected, dispatched, acknowledged, closed - four timestamps the system wrote. Nobody self-reports it and nobody can round it. That's why this is a number on the screen instead of a claim in a report."**

This is the answer to "how do we know it helps?" that does not depend on trusting anyone. The dashboard shows the median and breaks it into its three stages, so you can see *where* the time goes - detection is fast, and any delay is human.

## 10 IN-DEPTH ANALYSIS - /INSIGHTS

**The second tab: rankings, case mix, and how each illness is moving.**

> SAY THIS **"There's a second screen behind this one - rankings, the case mix, how each illness is moving over time. Every row opens into the actual encounters behind it."**

At a booth, mention it in one sentence and only open it if they ask. The live board is the story; this is the depth behind it. Every table row expands to the underlying records, and there is a table view of every chart for anyone who cannot read colour.

## 11 MEDICINE STALL - /STALL

**A second, genuinely independent signal. PIN 4444.**

> SAY THIS **"Medicine stalls tally what they sell - rehydration salts, zinc, antidiarrheals. Different reporter, different act (a purchase, not a consultation), different failure modes. Two independent signals agreeing is evidence. One signal shouting twice isn't."**

Two things make this more than a nice extra:

- **It leads.** People self-medicate before they consult, so sales rise before clinic visits do.
- **It sees the invisible.** It catches people who never reach a clinic at all - the single biggest blind spot in clinic-only surveillance.

Volunteer the caveat before they ask: **"In this demo both streams are simulated and one command writes both. You're seeing the architecture, not the evidence for it."**

If they ask what happens when the two disagree, that is the more interesting case: clinic cases up with no rehydration spike suggests it may not be waterborne, which changes where you send the crew first.

## 12 THE REFERRAL CARD

**A printed QR that carries the record itself, not a link.**

> SAY THIS **"A pilgrim found to have high blood pressure shouldn't lose that finding on the train home. The QR carries the record itself, not a link - so a village clinic in another state reads it with no network, no app and no ID system. Paper for the patient, digital for the system."**

This is also the answer to the identity question (topic 30). The one thing a national ID would genuinely buy - a finding following the patient home - we solved with a piece of paper.

## 13 THE PILGRIM SCREEN - /PUBLIC

**What a pilgrim sees. No app, no login, three languages, and never a case count.**

> SAY THIS **"It shows which areas are affected and which illness is going around. It never shows a case count - because a number invites comparison between areas, comparison moves crowds, and crowd movement is what kills at these events."**

Reached by scanning a printed poster. The poster carries its own area in the QR, so most readers never choose anything. Three tabs, because a pilgrim standing in a crowd has exactly three questions: what is happening where I am, is anywhere else affected, and what should I be doing anyway.

**Three care levels, not seven.** The official's map is a seven-step heat ramp keyed to case counts. The pilgrim map keeps the shape and swaps the quantity - three levels keyed to what to *do*, because there are only three things to do. A count ramp with the digits removed is still a count ramp; a reader infers the number from the colour.

| LEVEL | WHEN | WHAT IT MEANS |
|---|---|---|
| **Take extra care** | an act-tier alert - a crew was sent | Red tile, illness named, advice targeted at whoever is falling ill. |
| **Take care** | watch-tier - being observed | Amber tile, illness named. |
| **All clear** | nothing flagged | Plain tile. |

The level is **the same tier the health official sees**, so the two screens can never contradict each other. The grid stays in fixed order rather than sorted worst-first, because a leaderboard is an instruction to move.

**Targeted advice.** When one vulnerable group is over-represented, the page shows advice written for that group - keep feeding the child, ORS is free, and the one symptom that means go now. The targeting is **share-based, not count-based**: 18-39 is the largest band at any festival, so "most cases" would point there almost every time and tell a parent nothing.

Each affected area also names the illness and lists what it looks like and how it spreads - because "more stomach illness here" only helps if you can recognise it starting in the person beside you.

> BE STRAIGHT **If asked about the languages: "The Hindi and Marathi haven't been verified by a native speaker, and the symptom wording follows standard public-health guidance but hasn't been reviewed by a clinician. Both are required before this is shown to a real pilgrim, and both are noted in the source file."**

PART C - THE DETECTION

## 14 CELLS AND WINDOWS

**Twelve sectors times six syndromes = 72 cells, each compared against its own past at the same time of day.**

> SAY THIS **"We cut the festival into seventy-two cells - twelve sectors by six illness types - and score each one every five minutes against its own history at the same time of day."**

The current window is the **last 6 hours**. The baseline is the same 6-hour slot on each of the previous three days: 24, 48 and 72 hours back.

**Why the same time of day?** Clinics are busy in the morning and quiet at 3am. Averaged against the night windows, **every single afternoon looks like an outbreak**. Comparing an afternoon only with previous afternoons removes the daily rhythm without having to model it.

**Why compare a sector only with itself?** A busy sector is allowed to be busy. We are not asking "is this more than elsewhere", we are asking "is this more than here normally is".

## 15 THE FORMULA

**One line of arithmetic, deliberately small enough to check by hand.**

> SAY THIS **"Observed minus the mean, over the max of the standard deviation, the square root of the mean, and one. It fires at three, with a minimum of eight cases."**

Written out: **z = (observed - mean) / max( standard deviation, sqrt(mean), 1 )**

| TERM | WHY IT IS THERE |
|---|---|
| **observed - mean** | How far above its own normal this cell is. |
| **standard deviation** | A cell that swings wildly day to day needs a bigger jump to be surprising. A steady cell does not. |
| **sqrt(mean)** | The Poisson floor. See the next topic - this is the one worth explaining. |
| **the final 1** | Covers a mean of zero, where even sqrt(mean) is zero. Prevents division by zero. |

## 16 THE POISSON FLOOR

**The single term that stops the detector screaming, and the best thing to explain if they are technical.**

> SAY THIS **"Three quiet days in a row give you a standard deviation of zero - and then z goes infinite and every cell screams. Count data has irreducible noise of about the square root of the mean, so we never let the divisor drop below that. That one term kills most small-sample false alarms."**

This is a good thing to volunteer, because it shows you understood the failure mode rather than copying a formula. If a sector saw 2, 2 and 2 cases on three days, the standard deviation is 0 - and 3 cases today would score as infinitely surprising. With the floor, the divisor becomes sqrt(2), and 3 cases scores about 0.7. Correctly boring.

## 17 THRESHOLDS AND TIERS

**Fires at z >= 3 and at least 8 cases. Act tier at z >= 4.**

> SAY THIS **"Three noise-widths and a minimum of eight cases. Below four it appears on the dashboard and sends nobody; at four or above it dispatches a crew. We report the two separately because they cost completely different things."**

**Why the minimum of eight?** Without it, one case against a baseline of zero scores infinitely. Eight is the smallest count worth a human walking somewhere.

**Why three?** Lower catches marginally more and buries the dashboard in noise. Higher saves few false alarms and costs hours of detection lag. Three is where the curve bends - and the harness sweeps the threshold, so we can show what a different one would have cost rather than asserting this one is right.

> CAREFUL **If they say "three sigma, so a one-in-740 event?" - do NOT say yes. "It's a triage threshold, not a p-value. It means this cell is at least three Poisson noise-widths above what the same six-hour slot looked like on the last three days. It's tuned to put roughly the right number of things in front of a human - which is exactly why we measured the false-alarm rate instead of quoting a tail probability."**

## 18 DEDUP AND COOLDOWN

**One outbreak raises one alert, and a resolved cell stays quiet for six hours.**

> SAY THIS **"An outbreak that spans four windows is one alert, not four. And once a crew closes an alert, that cell stays quiet for six hours - otherwise the very next scan raises it again on the same data, seconds later."**

Without dedup, a single outbreak would appear as six separate alarms and our measured numbers would flatter us badly - which is why the evaluation harness mirrors this behaviour exactly (topic 20).

The cooldown is anchored to when the alert was *resolved*, not when it was raised. The demo injector deliberately bypasses the cooldown for the sector it fires into, so the demo beat always works.

## 19 BASELINE POISONING - A REAL WEAKNESS

**One genuinely noisy historical day makes a cell partially deaf for 72 hours.**

> SAY THIS **"Here's a real weakness we found by breaking it ourselves. If a sector genuinely had a huge day three days ago, that day is in its baseline - so its mean and its spread both rise, and it goes partially deaf. We poisoned our own sector 5 with repeated demo injections until it needed about two hundred and ninety-five cases before it would say anything."**

Say this on purpose. **A limitation you found yourself and can reproduce on demand is far stronger evidence of rigour than one you were told about.** It is written into our own unit tests as accepted behaviour, and the recovery procedure is in the repository.

The real fix, if asked: a longer baseline with outlier trimming, or a rolling median instead of a mean. We did not build it because it trades detection speed for robustness and we had no real data to tune that trade against.

PART D - THE EVIDENCE

## 20 THE HARNESS

**A flight simulator for the detector. This is your strongest card.**

SAY THIS **"We didn't want to assert that it works, so we measured it. A thousand synthetic festivals, an outbreak injected at a random sector and a random hour, running the deployed detection function over every hour of each one."**

Each trial builds 168 hours - 72 of warm-up so the baseline has history, plus 96 actually evaluated. Hourly counts are Poisson draws around a rate set by the number of clinics, the syndrome mix and the time of day. Outbreaks grow exponentially from a point source.

**Four design choices make it a measurement rather than a demo. Know all four:**

- **It imports the deployed algorithm.** The same `scoreCell()` the live Worker calls - not a reimplementation that could quietly drift. Change the detector and these numbers change.
- **It is deterministic.** Seeded, so `--seed 42` reproduces identically on their machine. That is the difference between a measurement and an anecdote.
- **It mirrors the engine's dedup.** Without this one outbreak would count as six catches.
- **The false-alarm half has nothing injected.** It cannot be gamed by construction - we cannot accidentally tune ourselves to detect an outbreak that is not there.

DO THIS **Offer to run it: "cd worker && node eval/harness.mjs". Almost nobody can do this in front of a judge, and offering is worth nearly as much as running it.**

## 21 WHAT THE HARNESS PROVES

**A typical waterborne outbreak: caught every time, median 7 hours, at 8 cases instead of 33.**

SAY THIS **"A typical waterborne outbreak is caught every time, a median of seven hours after the first case, at eight cases - against thirty-three before that cluster would be obvious to someone reading a chart."**

| OUTBREAK SPEED | CAUGHT | MEDIAN LAG | CASES WHEN WE ALERT | CASES IF UNAIDED |
|---|---|---|---|---|
| tiny (~1 case/hour) | 98% | 15h | **9** | 26 |
| small, self-limiting | 100% | 9h | **8** | 48 |
| slow (doubles / 6h) | 100% | 10h | **9** | 41 |
| **typical (doubles / 4h)** | **100%** | **7h** | **8** | **33** |
| fast (doubles / 2.5h) | 100% | 5h | **8** | 28 |

*"Cases if unaided" is the counterfactual: the size the cluster reaches before it is obvious by eye - a 6-hour count at three times its own baseline, and at least 25 cases. Medians, not means, because outbreak latency has a long tail and a mean would flatter us.*

## 22 FALSE ALARMS

**1.5 a day dispatch a crew. 5.4 a day are dashboard-only.**

SAY THIS **"One and a half a day dispatch a crew. Five point four a day appear on the dashboard and send nobody. We report them separately because they cost completely different things - one costs attention, the other costs a vehicle and four people."**

These come from the clean half of the evaluation, where nothing is injected at all.

VOLUNTEER THIS **"Those figures come from traffic generated as a pure Poisson draw. Our own live simulator is burstier than that, and against it the detector fires considerably more often. Neither model is wrong - they describe different worlds. The honest claim is 6.8 a day under a Poisson model, more under bursty traffic, and the real rate depends on how bursty real festival traffic is."**

## 23 THE SIMULATOR

**Fake data through a real pipe. It has no special access.**

> SAY THIS **"It's synthetic, written through the same public API a real tablet writes to. The detector can't tell the difference, because there is no difference to tell. That's exactly what let us measure across a thousand festivals instead of asserting it once."**

The rule that makes this defensible is in our project spec: **the simulator must write through the same API as the real app.** No direct database connection, no privileged key.

| COMMAND | WHAT IT DOES |
|---|---|
| **simulate.js backfill** | ~4,900 encounters over 84 hours with the real daily rhythm. Needed after any baseline reset - the detector needs history to compare against. |
| **simulate.js live** | A trickle of new patients so the board is never a dead screen. |
| **simulate.js outbreak 5** | A point-source waterborne cluster, growing wave by wave. |
| **simulate.js pharmacy** | The independent medicine-sales stream. |

Traffic shape: six patients per clinic per peak hour, times a time-of-day factor, times a random 0.6 to 1.4. Age bands follow a realistic festival mix (5% under-five, 25% over-sixty).

**Say "your data is fake" before they do.** Volunteering it costs nothing and buys the room. Being caught on it costs everything.

### PART E - THE ENGINEERING

## 24 ARCHITECTURE

**Static assets and two Cloudflare Workers. No server of ours to keep alive.**

> SAY THIS **"There's no backend of ours. The app is static assets on Cloudflare, the data is Supabase, and the detection engine is a second Worker with two cron triggers. Nothing to scale by hand on a bathing day."**

| LAYER | CHOICE | WHY |
|---|---|---|
| **Frontend** | React 19 + Vite, code-split lazy routes | No UI framework, no CSS framework, no state library. The pilgrim page is a few KB because it opens on cheap phones on bad networks. |
| **Data** | Supabase - Postgres, PostgREST, Realtime | The dashboard subscribes to changes rather than polling. |
| **Engine** | Cloudflare Worker, zero dependencies | Two crons: detection every 5 minutes, live traffic every minute. |
| **Hosting** | Cloudflare Workers static assets | Nothing of ours to keep alive during a festival. |
| **Auth** | PIN codes - deliberately not real auth | Demo theatre. A real deployment uses the health department's own identity, not ours. |

## 25 THE DATABASE

**Nine tables, row-level security on all of them, and the public key cannot delete.**

> SAY THIS **"Nine tables with row-level security on every one. The key that ships in the browser can read, insert and update - it cannot delete anything. Cleanup has to be run by hand by someone with the real credentials."**

The tables: clinics, encounters, baselines, alerts, tasks, referrals, pharmacy_sales, pharmacy_signals, pharmacy_stalls.

If they ask about the key being public: it ships inside the JavaScript bundle, so it is public by construction - hiding it would be theatre. What protects the data is the row-level security policy, not the secrecy of the key. That is how Supabase is designed to be used.

## 26 PRIVACY

**We never collect anything that identifies a patient.**

> SAY THIS **"We store an age band, sex, symptoms, diagnosis and medicines. No name, no phone number, no ID number, no location finer than the sector. There is nothing in this database that identifies a person."**

This is a strong point and it is easy to forget to make. It is also not an accident - the detector counts cases per sector, so identity would add nothing (topic 30). The pilgrim screen records nothing at all: no login, no account, no tracking.

The referral card is the one place a record travels with a person, and it travels **in their pocket**, on paper they hold - not in a database they cannot see.

## 27 CONFIG-DRIVEN

**Point it at a different gathering by editing one file.**

> SAY THIS **"Sectors, clinics, syndromes, age bands and thresholds are all config. Pointing this at a different mass gathering means editing one file, not rewriting the system. It is not hardcoded to Nashik."**

Judges assessing whether a project could be real ask this in some form almost every time, and it never comes up in a demo unless you raise it. Raise it.

### PART F - THE HARD QUESTIONS

## 28 WHY NO AI

**Explainability is an adoption feature, and there is nothing for a model to learn.**

> SAY THIS **"Deliberately. A district health officer will act on 'fourteen against four expected' and will ignore a black box - so explainability here is an adoption feature, not a limitation. And practically: seventy-two series with no labelled outbreak history gives a model nothing to learn that arithmetic doesn't already give you."**

If they push: we would need a history of labelled outbreaks at mass gatherings to train on, and that data does not exist. Any model would be learning our own simulator's assumptions and reporting them back to us as a result.

## 29 THE BATHING DAY PROBLEM

**Our sharpest weakness. Concede it immediately.**

> SAY THIS **"Correct, and it's the first thing we'd fix. We detect on raw counts, so a genuine crowd surge would false-positive across every sector at once. The fix is rates per thousand people present, using the crowd estimates the administration already produces. It's designed and deliberately not built - we chose to make what exists solid rather than half-build a second thing."**

**Do not defend this one.** Conceding instantly and having the fix ready is worth more than any argument, and judges who ask it are usually testing whether you know your own limits.

## 30 AADHAAR AND ABHA

**Identity adds nothing to what we do, and it would cost us the ten-second entry.**

> SAY THIS **"Aadhaar doesn't hold medical history - authentication returns demographics. So scanning it would give us a name we deliberately don't want, and none of the clinical history you're imagining. ABHA is the system that holds records."**

Then the real answer: **identity adds nothing to detection.** We count cases per sector against that sector's own history, and a z-score does not care who anyone is. Adding identity would cost us the ten-second entry, and at a Kumbh it would turn away exactly the elderly and rural pilgrims whose fingerprints fail authentication - the people most at risk.

There is also a legal answer if they know the domain: since the 2025 rules a private entity must route each use case through a Union or State ministry to UIDAI. That is a governmental process, not an integration.

And the constructive close: **"If this went real, ABHA is the right rail and it already exists - consent-based record linking, with a public sandbox. We'd attach to that rather than to Aadhaar. But it changes nothing about detection."**

## 31 WHAT IS NOT BUILT

**Say these before they find them.**

SAY THIS **"Crowd normalisation, a clinician review of the advice wording, native-speaker verification of the Hindi and Marathi, and validation against the real Jana Swasthya dataset. All four are written down. None are hidden."**

| AREA | STATUS |
|---|---|
| **Crowd normalisation** | Designed, not built. Topic 29. |
| **Real-world validation** | Not done. Every number is from synthetic data. The highest-value next step is the public Jana Swasthya dataset on Harvard Dataverse. |
| **Clinical review** | Advice wording and the antibiotic table both need a clinician. |
| **Translations** | Not verified by native speakers. |
| **Authentication** | PINs only, by design for a demo. |
| **Baseline poisoning** | Known, documented, reproducible. Topic 19. |

The framing that makes this a strength: **"We'd rather ship four solid screens than eight half-finished ones."**

## 32 EVERY NUMBER AND ITS SOURCE

**If a number is not on this list, do not say it.**

SAY THIS **"Everything we quote is either measured by our own harness, which anyone can rerun, or cited to a published paper."**

| FIGURE | MEANING | SOURCE |
|---|---|---|
| **8 vs 33 cases** | Cluster size when we alert vs when it becomes obvious by eye. | worker/eval/RESULTS.txt |
| **7 hours** | Median detection lag, typical outbreak. | Same run. |
| **100% caught** | Typical, fast, slow, self-limiting. Tiny: 98%. | Same run. |
| **1.5 / day** | Act-tier false alarms - a crew goes out. | Clean festivals. See topic 22 caveat. |
| **5.4 / day** | Watch-tier - dashboard only. | Same. |
| **2.2 seconds** | Time to log one patient. | Timed on the deployed app. |
| **31%** | Antibiotic prescribing at the 2015 Kumbh. | Shaikh et al., JMIR Public Health and Surveillance, 2023 |
| **2,000+ visits** | Jana Swasthya day one, 30+ clinics. | Harvard Mittal South Asia Institute |

NEVER SAY **Never quote "600 cases reduced to 35". It is a logistic curve fitted to nothing, and it would sit in the same conversation as the measured 33-to-8. Two counterfactuals an order of magnitude apart invites the one question you cannot answer well.**

If you get one sentence, make it this one: **"Every clinic treats its own patients well, and nobody compares notes. We do that every five minutes - and in testing we catch a typical outbreak at eight cases instead of thirty-three, about seven hours earlier."**